

Clinician preferences for explainable AI in critical care: a comparative study of interpretable models and visualizations for intubation decision support

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ABSTRACT

Background:: The complexity of many AI models hinders their clinical adoption because the clinicians using them do not regard them as transparent. This study addresses the lack of clinician-centered explainable AI (XAI) interfaces by designing and evaluating intuitive visual explanations for intubation prediction, testing the hypothesis that workflow-compatible designs enhance acceptance.

Objective:: This study compares three, time-aware, visual explanations for XAI-based intubation prediction and evaluate their acceptance, comprehension, and perceived utility among clinicians.

Methods:: We developed machine learning models to estimate the near-term risk of deterioration in the patient's condition which may lead to mechanical intubation using ICU time-series data. We generated global and local explanations using SHAP and designed three customized visual formats—a temporal force plot, a temporal bar chart, and a dual-encoded SHAP heatmap. Clinicians ($n = 206$) evaluated comprehension and usability using objective questions and a Likert-based survey.

Results:: Based on 4608 critically ill patients with 10 medical variables over 7 hours of data for each patient, the Random Forest (RF) model achieved the highest area under the curve (AUC): 0.94. Furthermore, the local explanations were customized and evaluated by 206 clinicians through a survey conducted on the Prolific platform. A customized heatmap representation was selected as the visualization with the highest perceived clinical utility and alignment with clinical workflows.

Discussion:: The reported findings support the need for explanation formats to be tailored to clinical reasoning and task context, supporting the concept of cognitive fit. The heatmap's close alignment with clinicians' mental models and its graphical integrity enhances interpretability and trust. This study demonstrates that explanation effectiveness depends on contextual relevance, rather than a universal standard, and that the presentation format itself significantly shapes clinicians' trust in XAI systems.

Conclusion:: This study advances clinical XAI by introducing a time-aware explanation framework for ICU intubation decisions. By integrating temporal trends with model reasoning, our visualizations closely align with clinicians' cognitive workflows. Rigorous clinician-centered evaluation identified the dual-encoded SHAP heatmap as the most useful and workflow-compatible visualization, highlighting the importance of explanation design alongside predictive accuracy for clinical adoption.

1. Introduction

ICU patients deteriorate unpredictably, and early recognition of respiratory failure is clinically critical. However, AI tools for predicting intubation remain limited by the opacity of many models, which affects

clinician trust and adoption [1–4]. Explainable AI (XAI) methods can help bridge this gap by providing an interpretable rationale for predictions, enabling clinicians to validate recommendations against the patient's recent physiological trajectory [5–8]. In practice, intubation decisions are rarely binary or instantaneous but are made through

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anticipatory assessment of evolving respiratory and physiological trends.

Despite the promising opportunities presented by XAI, there is one key limitation in current research concerning the evaluation of the explainability of AI models. While previous research has developed and validated the accuracy of different AI models before proceeding with the development and application of XAI models [9–11], few studies directly involve clinicians in evaluating XAI visualizations for critical care [12,13]. This is particularly problematic, given that clinicians are the direct recipients of the outcomes from the AI and XAI models, using their advice to make important clinical decisions. Therefore they should be involved in the design of visualizations which help them to understand the models within their clinical practice.

In this paper, we address this limitation in current research. We have developed multiple machine learning (ML) models to estimate the risk of deterioration and hence the probability of mechanical intubation for critical care patients using time-series data. We selected the best ML model based on its AUC (area under the curve) values [14]. Then, we developed and validated an XAI model based on SHAP (SHapley Additive exPlanations) values to estimate the feature contribution towards the AI prediction [15]. Finally, and most importantly, we conducted a survey of clinicians to evaluate three ways of visualizing the XAI, namely, via a heatmap, bar chart, and a force plot. Our analysis of the survey results shows clearly that clinicians prefer the heatmap as having the highest clinical utility and alignment with clinical workflows. Our approach to developing and evaluating XAI models offers clear benefits to deploying XAI in ICU units, including more transparency into AI prediction for intubation decisions, as well as alignment to clinician preferences for the use of such models in clinical practice.

2. Materials and methods

2.1. Method overview

In this study, intubation prediction is formulated as a near-term risk estimation problem. Using routinely collected ICU time-series data, the model estimates the probability that a patient's condition will deteriorate and hence that the patient will require mechanical intubation within the next hour, conditional on their physiological trajectory over the preceding 6 hours. The end of this 6-hour window represents the clinical “now” corresponding to the moment at which a clinician would reassess risk and consider escalation of respiratory support. Intubation events used for outcome labelling occur after the observation window, ensuring that the model is not informed by post-decision data. This framing reflects anticipatory ICU decision-making rather than immediate procedural guidance.

To achieve this, we first selected clinically relevant medical variables from the MIMIC-III database, which are crucial in intubation decisions. They are processed to have the standard format. Secondly, predictive ML models and an XAI model are developed to generate suggestions on whether the patients need intubation within the next hour. Thirdly, the explanation plots are customized to optimize the conveyance of information. In response to the identified needs of the end users, a survey was employed to enumerate and evaluate clinicians' acceptance and attitude towards this AI-assisted tool.

2.2. Data processing

We extracted data from the MIMIC-III database [16]. We included adult patients (≥ 18 years) with an ICU stay exceeding 7 hours to ensure data stability, particularly for low-frequency measurements (such as PO_2) [17]. To address class imbalance, we maintained a 1:2 ratio of intubated to non-intubated patients, yielding a final cohort of 4,608 individuals (see Fig. S1 and Table S1 in the Supplemental Information) [18]. Ten key medical variables were selected by clinicians as model inputs [19]. The selected medical variables were structured into a

sequential format, with each record tagged by a timestamp at hourly intervals. Data were processed into hourly medians for frequent measurements [20], with missing values imputed via the Last Observation Carried Forward (LOCF) method or population means [21] (see Table S2 for variables in the Supplemental Information).

2.3. Model development

We developed multiple ML models to classify time-series data and predict whether the patients need intubation within the next hour, namely, Logistic regression (LR) [22], Decision trees (DT) [23], Random Forest (RF) [24], eXtreme Gradient Boosting (XGBoost) [25], Explainable Boosting Machine (EBM) [26], Long Short-Term Memory (LSTM) [27], and Convolutional neural network (CNN) [28]. The optimal model was selected according to the highest AUC in the test dataset [14].

Of the various interpretability methods available, SHAP [15] was selected for its rigorous theoretical foundation and ability to provide both global and local model explanations. Unlike other approaches, such as LIME [6], which offers local approximations but may lack consistency, or permutation-based feature importance [29], which only delivers overall rankings, SHAP unifies these perspectives under a game-theoretically optimal framework [30]. It assigns each feature an importance value for every individual prediction, ensuring consistency and comparability across samples. This is particularly valuable in medical contexts, where understanding the rationale behind a specific decision—not just the model's general behavior—is critical for clinical trust and actionable insights [31,32]. Additionally, SHAP's compatibility with any ML model enhances its practicality, allowing researchers to interpret complex ensembles or deep learning systems without sacrificing performance [11,33]. These advantages make SHAP especially suitable for high-stakes healthcare applications, where transparency, accountability, and nuanced interpretation are essential.

2.4. Explainability evaluation

The integration of effective visual explanations in clinical decision support systems hinges on their ability to align with the cognitive processes of clinicians [34]. Well-designed visualizations, such as saliency maps [35] or waterfall plots [15], offer a more intuitive alternative to numerical or textual explanations by reducing cognitive load and facilitating a holistic understanding. This intuitiveness is critical for building trust and enabling effective human-AI collaboration [36]. To validate efficacy, the field has adopted human-centric evaluation, including user studies measuring diagnostic performance [37], eye-tracking experiments to assess attentional guidance [38], and structured surveys quantifying perceived usefulness and trust [39].

Building on this foundation, we developed an interpretable AI framework for intubation recommendation that integrates a clinical suggestion, a predictive probability score, and a visual explanation. Since standard explanation plots (e.g., conventional force plots) often lack temporal context [40]—which is key in clinical assessment—we designed three novel, customized visualizations: a time-aware force plot, a temporal bar chart, and a clinical heatmap. Each displays both feature contributions for the current prediction and historical trends of critical variables [41,42]. As the relative effectiveness of these formats was unknown, we conducted a systematic evaluation to identify which one best supports clinicians' cognitive needs [43].

The evaluation was conducted via a structured online survey administered through Qualtrics and disseminated via the Prolific platform. As illustrated in Fig. S2, the survey included seven multiple-choice questions assessing the clarity of each visualization and Likert-scale questions, adapted from the Technology Acceptance Model 3 (TAM3) [44,45] (see Table S4 in the Supplemental Information), to assess clinicians' user intention. A final question gauged participants' preference for future use. By gathering feedback from clinicians, this study moves beyond theoretical design toward an evidence-based selection of the

most effective visual explanation for clinical decision-making.

TAM3 was selected to evaluate clinicians' preferences for XAI visualizations. TAM3 posits that technology adoption is primarily driven by Perceived Usefulness and Perceived Ease of Use. We conceptualize the different XAI visualizations as system characteristics influencing these perceptions. Applying this model allows for a precise analysis of how each visualization affects clinicians' cognitive assessment and acceptance intentions, thereby identifying the most appropriate design for fostering trust.

3. Results

3.1. Predictive model performance

The final dataset included 4,608 patients with 10 features recorded hourly over 7 hours, randomly split into training (80%), validation (10%), and test (10%) sets. Among the models evaluated, Random Forest achieved the highest test AUC of 0.94, slightly outperforming more complex deep learning architectures (Fig. 1). This suggests that for this specific task involving smaller-scale multivariate time series, the sophisticated temporal feature learning of deep models may not offer a distinct advantage over the robust performance of tree-based models [46,47]. A comprehensive performance report, including AUC, accuracy, specificity, sensitivity, and F1-score, is provided in Table S5 in the Supplemental Information. Ultimately, the superior performance and inherent interpretability of tree-based models led to the selection of Random Forest as the final model for generating subsequent explanations.

3.2. Explainable model performance

The SHAP analysis results are summarized in Fig. 2. Globally, the bee-swarm and bar plots (Fig. 2a-b) reveal that FiO2 at the current hour (FiO2_T6) was the most influential feature in the model's risk estimation, followed by heart rate (HR_T6) and earlier values of FiO2 and respiratory rate. This feature's importance aligns with clinical intuition,

underscoring the significance of respiratory and hemodynamic parameters. At the local level, the force plot for a sample patient (Fig. 2c) illustrates how specific values—such as $\text{FiO2_T6} = 100$ and $\text{HR_T6} = 123$ —contribute to elevating an individual patient's intubation probability to 0.88.

While standard explanation plots effectively illustrate feature importance at a single point in time, they often lack the capacity to represent temporal dynamics, thereby limiting their clinical utility for longitudinal data. To address this gap, we developed three novel, customized visualizations that explicitly integrate the temporal dimension of medical variables and their evolving impact on model predictions. As illustrated in Figs. 3–5, these include: (a) a time-aware force plot, which segments and color-codes Shapley values by time intervals to visually track contribution trends; (b) a temporal importance bar chart, which aggregates absolute contributions over selected windows to highlight persistently influential variables; and (c) a dual-encoding heatmap, which uses color intensity to represent Shapley values while overlying actual variable values as text, enabling simultaneous assessment of feature importance and clinical measurements. Critically, all three visualizations are designed not only to display the original variable values but also to reveal how each variable's influence on the prediction changes over time. This temporal explication allows clinicians to contextualize the AI's recommendation within the patient's recent clinical trajectory, bridging a crucial gap between static explanations and dynamic clinical reasoning.

3.3. Explainability evaluation

A total of 217 questionnaires were received via the Prolific platform. Following the removal of unqualified questionnaires that exhibited extreme outliers, insufficient response times, linear or regular answers, and other such anomalies, a total of 206 valid data points were obtained. The demographic information for the study sample is displayed in Table S6 in the Supplemental Information.

The primary objective of the questionnaire was to assess the clinicians' perception towards the visual explanations. Heatmaps were

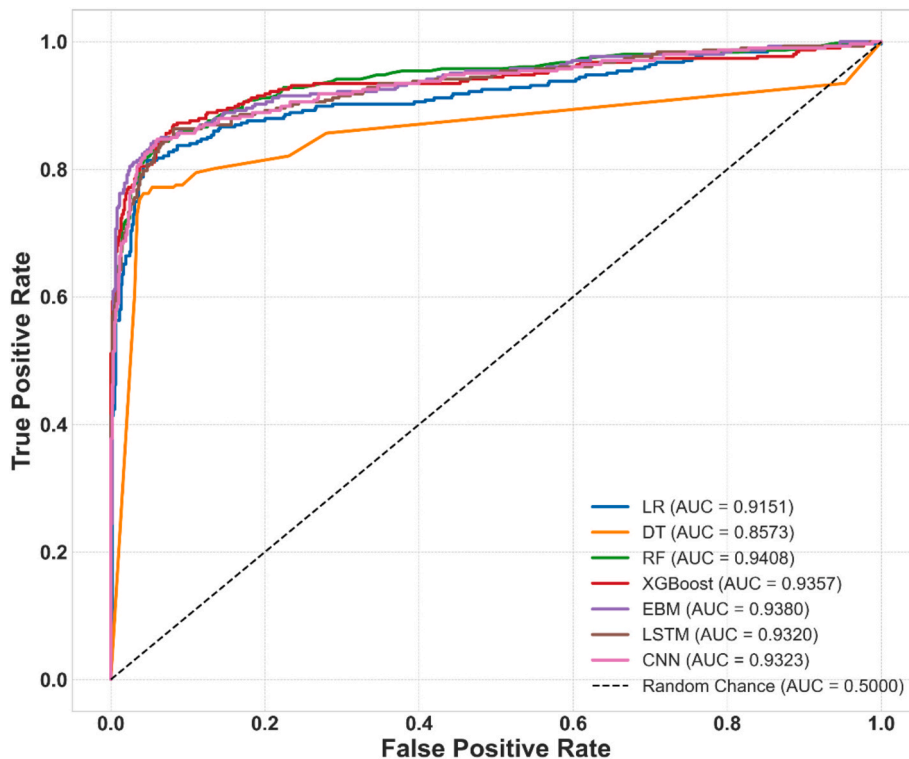


Fig. 1. Receiver operating characteristic (ROC) curves of the predictive models.

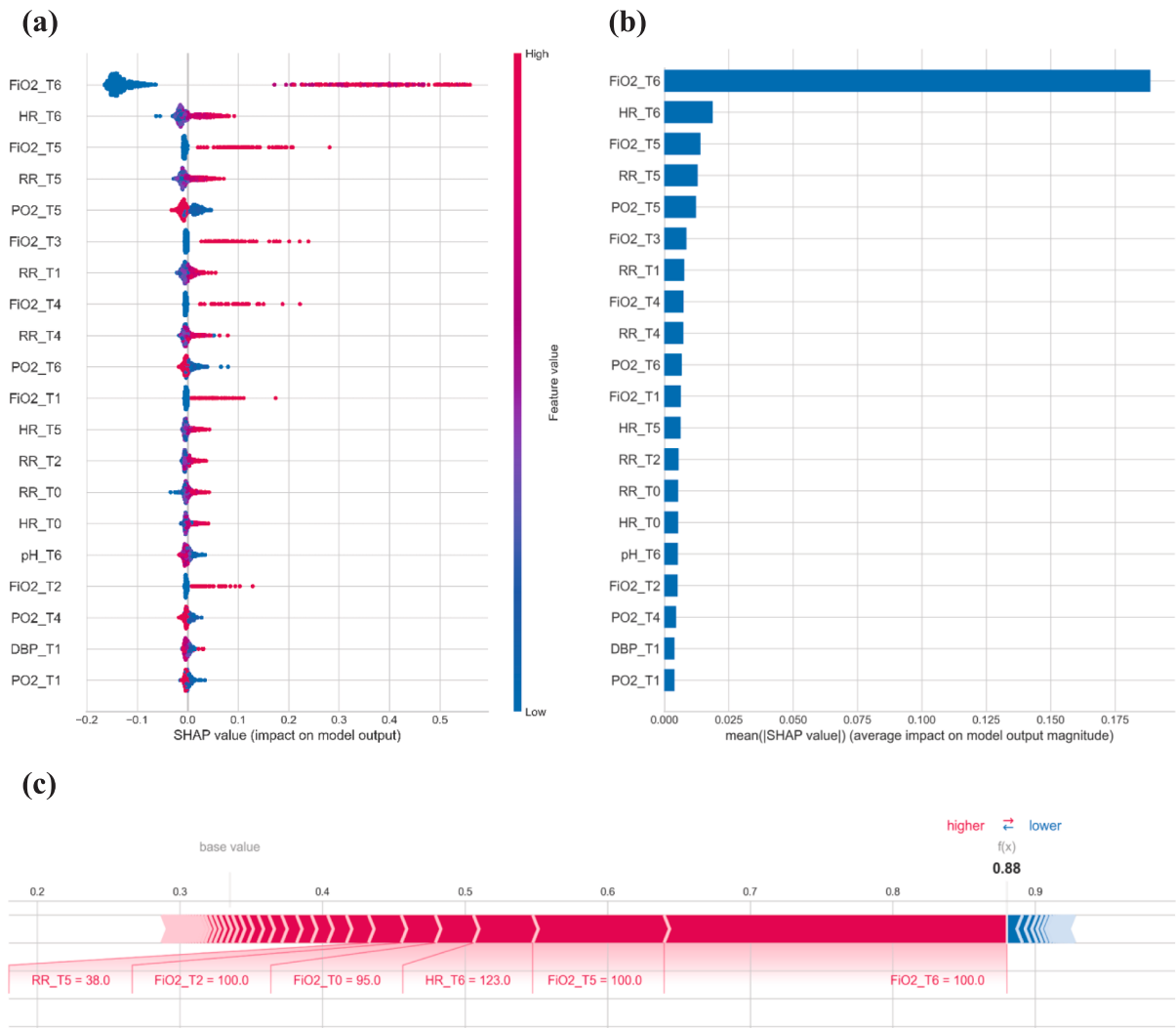


Fig. 2. (a) SHAP bee-swarm plot; (b) SHAP feature importance bar plot; (c) Force plot for an individual patient case.

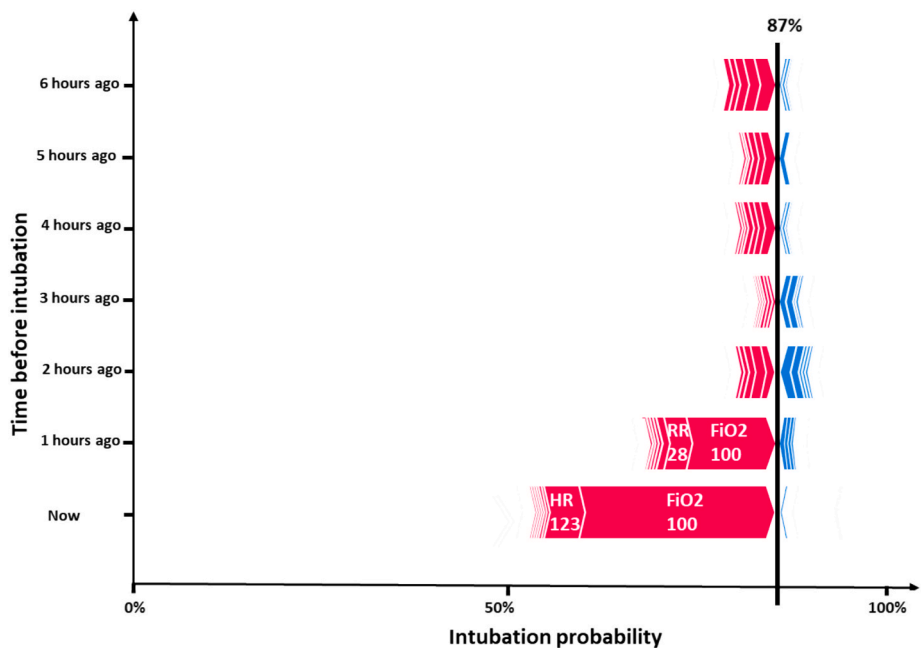


Fig. 3. A time-aware force plot.

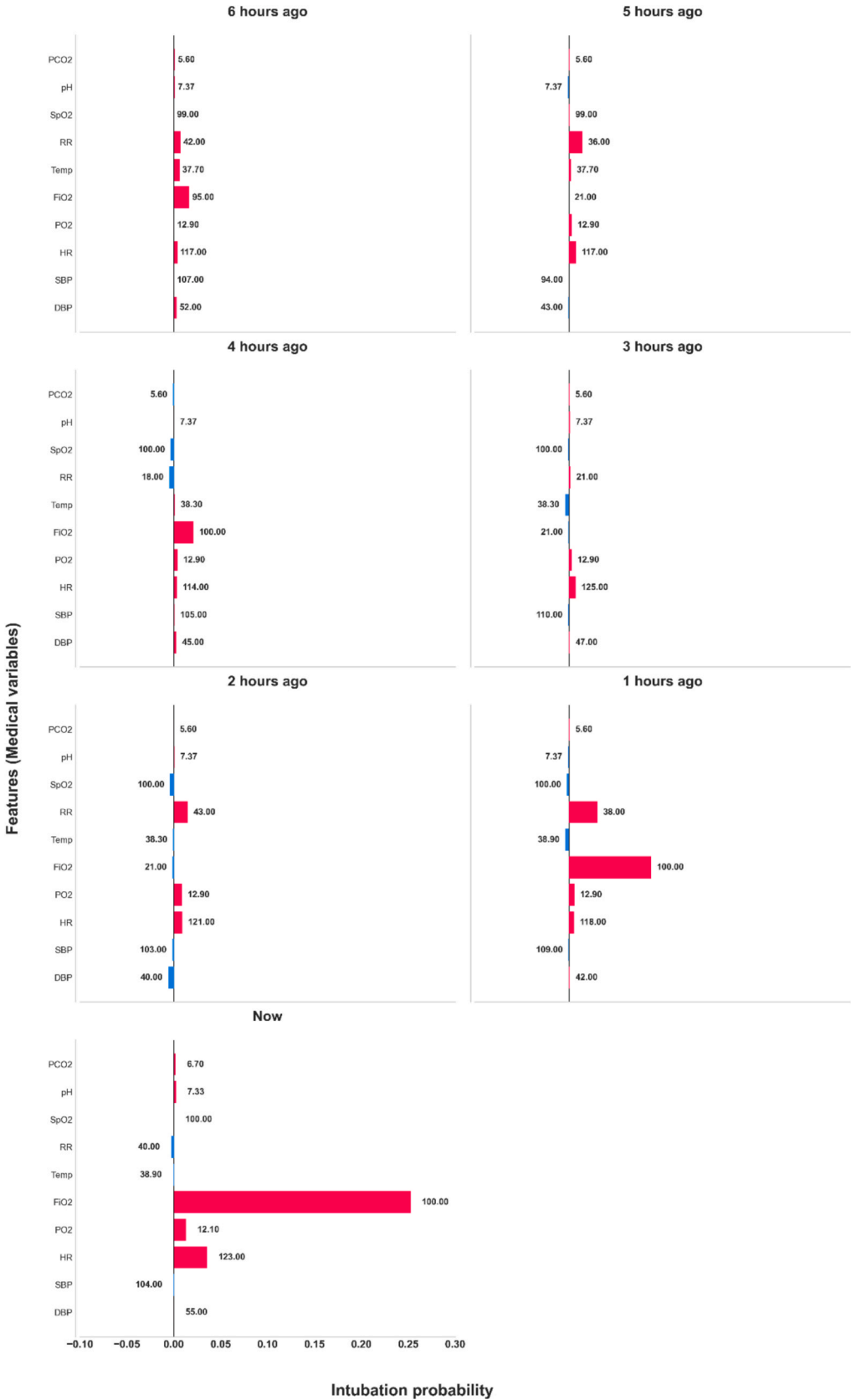


Fig. 4. A temporal importance bar chart.

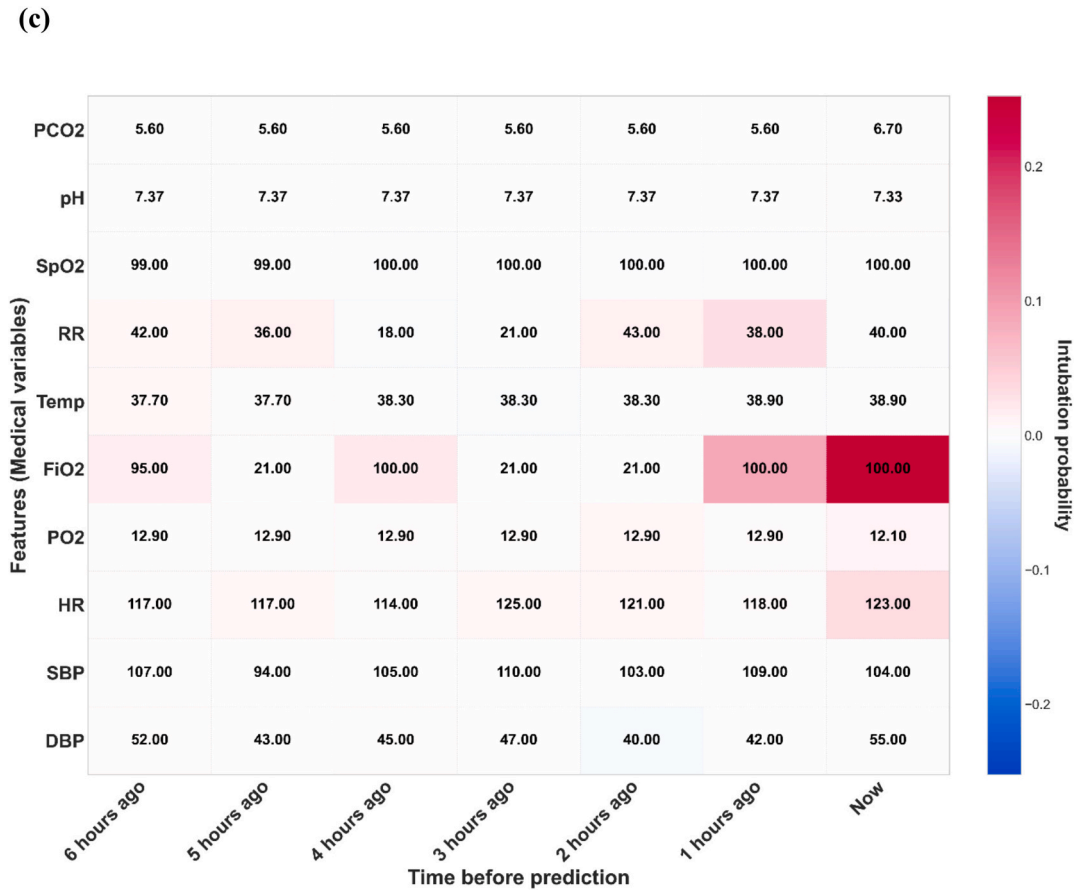


Fig. 5. A dual-encoding heatmap.

considered the most acceptable interpretable graph, with approximately 55% of participants choosing them. 30% participants prefer the force plot, and only 15% of the participants selected the bar chart.

The comprehension test (7 questions) achieved an average accuracy of 78% (see Table S3 for question stems and Fig. S3 for comprehension result in the Supplemental Information). Results show that Question 1 (denoted Q1), evaluating all three plots simultaneously, scored lowest (47%). Q2, Q4, and Q6—identical questions presented in different graphical formats—all achieved consistent accuracy (80%). In contrast, Q3, Q5, and Q7, derived from distinct graphics, varied in performance. Among the explanation formats, the heatmap scored lowest (76% accuracy), followed by the bar chart (90%), and the force plot (92%). While overall comprehension was satisfactory, explanation clarity differed: the force plot was perceived as clearest, whereas the heatmap performed poorest.

Analysis of the TAM3 survey results, conducted via a five-point Likert scale, provides deeper insight into user perception. The heatmap was rated highest across the core constructs of Perceived Ease of Use ($M = 3.84$, $SD = 1.13$), Perceived Usefulness ($M = 3.84$, $SD = 1.03$), and behavioral intention ($M = 3.79$, $SD = 1.08$), suggesting that despite its slightly lower comprehension score, clinicians who understood it found it the most practical and intuitive tool for interacting with the AI's recommendations. The measurement model analysis confirmed the reliability and validity of the survey instrument, with all criteria met (see Tables S7–S10 in the Supplemental Information).

Overall, the heatmap emerged as the best visual explanation for supporting intubation decisions. It performed best in clinician preference tests, achieving the highest ratings for perceived clinical utility and alignment with clinical workflows, despite the brief onboarding process. In contrast, the heatmap scored slightly lower in clarity compared to some alternatives. We do acknowledge, however, that these findings

remain constrained by limited user training and by interface features such as legends and tooltips, which could influence interactive usability and interpretation.

4. Discussion

This study developed an explainable AI approach and evaluated visual explanations to assist clinicians in an Intensive Care Unit (ICU) with intubation-related decision making. Clinicians make intubation decisions by integrating evolving physiological trends rather than isolated values [4]. Standard SHAP plots, while informative, do not convey this temporal context and therefore do not align well with real clinical reasoning. To address this limitation, we adapted SHAP-based explanations to include temporal information, creating three formats designed specifically for ICU use: a time-aware force plot, a temporal bar chart, and a dual-encoded heatmap. These formats allow clinicians to simultaneously interpret the patient's trajectory and understand how each variable influences the model's prediction. Importantly, the model is not intended to recommend intubation, but to highlight patients whose recent physiological trajectory resembles patterns that commonly precede intubation, thereby supporting early recognition of deterioration rather than procedural timing.

Our findings show that the heatmap was preferred not because it provided different information, but because it aligned more closely with how clinicians naturally synthesise ICU data. By combining raw values and model attributions in the same view, the heatmap supported rapid pattern recognition and reduced the mental steps needed to connect physiological trends with predicted risk [48]. This suggests that explanation formats must match clinicians' real decision-making processes, not just convey feature importance.

In contrast, both the force plot and the bar chart separated the

clinical data from the model's reasoning. This required clinicians to mentally reconnect the two, increasing cognitive effort and reducing perceived usability [49–52]. Although these formats were still understood, their structure made them less suitable for fast-paced environments such as the ICU. These forced clinicians to mentally reintegrate these disparate pieces to form a holistic understanding of the case, a process that demands working memory and is less efficient than the heatmap's integrated view.

These findings resonate strongly with established principles in visualization and human–computer interaction. They align with the concept of “cognitive fit” [53], which posits that problem-solving effectiveness depends on the match between the problem representation and the task [54,55]. The heatmap achieves a high degree of cognitive fit with the spatial, case-based mental models of clinicians. This supports the principle of “graphical integrity”, where the heatmap maintains a direct link to the original data context, maximizing data-ink ratio [56], whereas the other formats are re-encodings that introduce abstraction. Furthermore, this study empirically validates the core tenet of human-centered XAI that explanations must be tailored to the specific user and task context [57]. There is no universally optimal explanation; effectiveness is contingent. Our results also extend earlier work in clinical systems, showing that explanation presentation independently influences trust, separate from content [58].

The survey results reinforce that an explanation format must be both easy to interpret and perceived as clinically useful to support adoption [45]. Importantly, clinicians were more willing to use an AI tool when its explanations required minimal mental effort and clearly linked physiological trends to predicted risk; criteria most consistently met by the heatmap.

In addition, we acknowledge the limitations of the study. First, the evaluation was conducted outside the clinical environment, without the time pressures and competing priorities typical of ICU workflows. Second, although our sample included a broad mix of clinicians, only a minority were practicing ICU clinicians. Third, the model relied on a limited set of routinely available variables and a fixed 7-hour observation window, which may not fully capture the complexity of real-time ICU decision-making. Finally, we assessed perceived usefulness and understanding rather than actual changes in clinical performance.

Future work could further examine our models and findings in real ICU settings, ideally integrated into the electronic health record. Such studies could evaluate not only clinician acceptance but also decision quality, situational awareness, and potential unintended consequences [59,60]. Comparative studies across different patient populations and prediction tasks would further clarify the generalisability of these findings.

5. Conclusion

This study presents an advancement in clinical XAI by developing and validating a time-aware explanation framework for anticipatory intubation decision-making in the ICU. Our primary innovation lies in moving beyond static feature importance by designing visualizations that integrate temporal trends with model reasoning, thereby directly aligning with clinicians' cognitive workflows. The rigorous, clinician-centered evaluation—employing surveys, comprehensive tests, and preference questions—empirically established the dual-encoded SHAP heatmap as the visualization with the highest perceived clinical utility and alignment with clinical workflows. This underscores that for clinical adoption, explanation design is as critical as predictive accuracy.

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CRedit authorship contribution statement

Tiantian Xian: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Nikolay Mehandjiev:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Panos Constantinides:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Yu-wang Chen:** Writing – review & editing, Validation, Supervision, Software. **Qudamah Quboa:** Validation, Software. **Gareth Kitchen:** Writing – review & editing, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

MIMIC data files can be accessed via a documented process on the **MIMIC website**. The code is shared via GitHub (<https://github.com/Tina-Sheffield/Intubation-prediction-and-SHAP.git>).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmedinf.2026.106287>.

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